University of Algiers 1
Faculty of Science
Department of Computer Science

Academic Year 2025–2026

Lab Project: development of a Mini Compiler

The aim of this project is to develop a mini-compiler for a language called “Lang_F” that performs the different phases of compilation: lexical, syntax, and semantic.

I. Description of the Lang_F langague

II.1 The general structure

```
a sequence of declarations
BEGIN
a sequence of instructions
END
```

A) List of declarations

In this language, we can declare simple variables and constants.

Simple variable: **Type VariableName;**
Constant: **CONST Type ConstantName;**

Exemple

```
CONST INT tic=56;
INT HG_21, art, loop6;
FLOAT JK, L7, M66_0;
BOOL opinion;
```

- The language accepts three types : **INT**, **FLOAT**, and **BOOL**.
- A set of variables can be declared by separating the variable names with commas (,).

- Each **variable name** is an **identifier**. An **identifier (idf)** starts with an uppercase alphabetical letter (A–Z), followed by a sequence of digits (0–9), alphabetical letters (A–Z and a–z), or the symbol ‘_’. Its length must not exceed 10 characters.

  Identifiers must satisfy the following conditions:

  - The identifier must not contain successive _ *symbols*

- The _ symbol must not appear at the end of the identifier.

An identifier can be assigned a constant. There are two types of constants:

a) **Integer constants:** A sequence of digits, which may be signed or unsigned.
b) **Real constants:** A sequence of digits containing a decimal point.

B) Instructions

In this language, **only** the following instructions are allowed:

Instruction	Description	Exemple
Affectation	**Idf := expression arithmétique ;**	A :=C*B; A :=2;

Loop	**For (var :=valInit, condition, compteur) {
	Bloc Instructions
	}**
		For (I:=1 ;i<=10
		;i++) {
		B:=B+2;
		}

condition	**If (Condition) { Bloc instructions} else {
	Bloc instructions}**
		If (a==2) {
		B :=5 ;
		C :=B+1 ;
		}

- Arithmetic expressions use arithmetic operators: : + - *
- Conditions use relational operators: < <= <> == > >=

C) Comments

Comments are a sequence of characters enclosed between '/' and '/', and must not contain any '/' within them.

Tasks to be completed

Below are the different phases to implement in order to build the requested compiler.

- **Lexical analysis** using the **FLEX** tool.
- **Syntax analysis** using the **Bison** tool.
- **Semantic analysis** for the following errors:
    - Undeclared identifier
    - Identifier declared twice

    - Type incompatibility
    - Attempt to change the value of a constant

- **Symbol table management**
  The table must contain at least the following fields:
    - **Name:** the identifier (name of the variable or constant).
    - **Type:** the type of the variable or constant.
    - **Const:** indicates whether the identifier is a constant or not.

- **Error handling:**
  the developed system must display appropriate error messages at each stage of the compilation process, specifying the line number and column of the error.